(a) Shared spectral encoder

Theory y

Proxy (c)

Theory view

ỹ

x

Experiment

Spectrum Transformer

$$\text{Loss} = \frac{1}{2}(L_{Q\to R} + L_{R\to Q})$$

Shared weights

Projection
+ L2 norm
128 → 64

u

u

64-D unit vectors

Theory refs
96 spectra

Spectrum Transformer

Projection
+ L2 norm

v

Shared latent space

● queries ■ theory refs

Pair metadata

Canonical identity
Positive-pair mask

Source label
QME14S / fg26

(b) Multi-positive alignment

U

V

U → V / V → U

Identity matching

Same identity

Different identity

(c) Proxy augmentation

Theory

Fixed proxy

Random views

(d) Hard negatives

Real query

Embedding

Training ranks

Exclude positives

Sample one
Top-16 errors

(e) Coverage masking

Spectrum + mask

unmeasured

Valid-bin norm

Patch masking

Patch coverage weights

Weighted pooling